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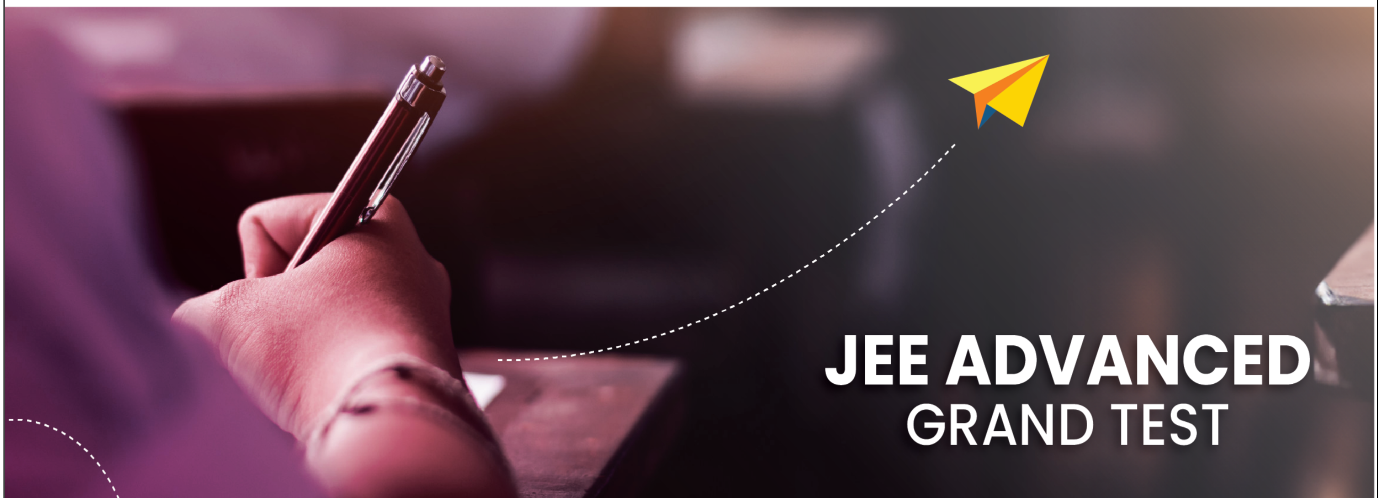

Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr.Super60_NUCLEUS_BT	**Paper -2(Adv-2021-P2-Model)**	**Date: 28-07-2024**
Time: 02.00Pm to 05.00Pm	**GTA-01**	**Max. Marks: 180**

28-07-2024_Sr.Super60_NUCLEUS_BT_Jee-Adv(2021-P2)_GTA-01_Syllabus

MATHEMATICS : TOTAL SYLLABUS

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

Name of the Student: ____________________ **H.T. NO:** ☐☐☐☐☐☐☐

JEE-ADVANCE-2021-P2-Model

Time: 3:00Hr's **IMPORTANT INSTRUCTIONS** Max Marks: 180

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 6)	Questions with Multiple Correct Choice with Partial mark	+4	-2	6	24
Sec – II(Q.N : 7 – 12)	Paragraph Questions with Numerical Value Answer Type	+2	0	6	12
Sec – III(Q.N : 13 – 16)	Paragraph Questions with Single Answer Type	+3	-1	4	12
Sec – IV(Q.N : 17 – 19)	Questions with Non-negative Integer Value Type	+4	0	3	12
Total				**19**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 20 – 25)	Questions with Multiple Correct Choice with Partial mark	+4	-2	6	24
Sec – II(Q.N : 26 – 31)	Paragraph Questions with Numerical Value Answer Type	+2	0	6	12
Sec – III(Q.N : 32 – 35)	Paragraph Questions with Single Answer Type	+3	-1	4	12
Sec – IV(Q.N : 36– 38)	Questions with Non-negative Integer Value Type	+4	0	3	12
Total				**19**	**60**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 39 – 44)	Questions with Multiple Correct Choice with Partial mark	+4	-2	6	24
Sec – II(Q.N : 45 – 50)	Paragraph Questions with Numerical Value Answer Type	+2	0	6	12
Sec – III(Q.N : 51 – 54)	Paragraph Questions with Single Answer Type	+3	-1	4	12
Sec – IV(Q.N : 55 – 57)	Questions with Non-negative Integer Value Type	+4	0	3	12
Total				**19**	**60**

PHYSICS

Max. Marks: 60

SECTION-1(Maximum Marks: 24)

One or More Type

- This section contains SIX (06) questions.
- Each question has FOUR options (A), (B), (C) and (D). ONE OR MORE THAN ONE of these four option(s)
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:

  Full Marks: +4 If only (all) the correct option(s) is(are) chosen;
  Partial Marks: +3 If all the four options are correct but ONLY three options are chosen;
  Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
  Partial Marks: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
  Zero Marks: 0 If unanswered;
  Negative Marks: −2 In all other cases.

1. A ray of light travelling in the direction $\frac{1}{5}(3\hat{i}+4\hat{j})$ is incident on a plane mirror. After reflection, it travel along the direction $\frac{1}{5}(3\hat{i}-4\hat{j})$. Then choose the correct statement(s).

A) Mirror is kept in x-y plane

B) Normal of the mirror is along $-\hat{j}$

C) Angle of incidence is 37°

D) Mirror is kept in x-z plane

2. A flat plate is moving normal to its plane through a gas under the action of a constant force F. The gas is kept at a very low pressure. The speed of the plate v is much less than the average speed u of the gas molecules. Which of the following options is/are incorrect? (Assume e=1 and molecules strike normally to the plate)

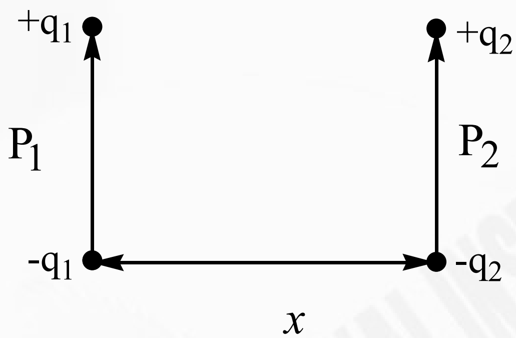
A) The resistive force experienced by the plate is proportional to v

B) The pressure difference between the leading and trailing faces of the plate is proportional to uv.

C) The plate will continue to move with the constant non-zero acceleration, at all times

D) At a later time the external force F balances the resistive force.

3. Figure shows two small electric dipoles of dipole moments P_1 and P_2 parallel to each other and placed at a distance x apart. 'x' is very large, compare to the separation between charges then:



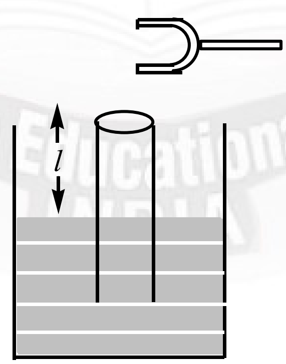
A) they will repel each other

B) they will attract each other

C) Force of interaction between dipoles is of magnitude of $\dfrac{3P_1P_2}{4\pi\varepsilon_0 x^4}$

D) Force of interaction between dipoles is of magnitude of $\dfrac{6P_1P_2}{4\pi\varepsilon_0 x^4}$

4. A tuning fork of known frequency is held at the open end of a long tube, which is dipped into water as shown in the figure. The tuning fork of frequency 165 Hz resonates in first and second harmonic when air columns have lengths $l_1 = 50 \pm 0.5cm$ and $l_2 = 150 \pm 0.1cm$. Then



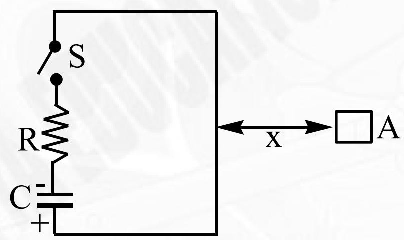
A) Speed of sound in air is $(330 \pm 1.98) m/s$

B) Speed of sound in air is $(330 \pm 1.2) m/s$

C) Percentage maximum error in speed of sound is 0.6%

D) Percentage maximum error in speed of sound is 0.4%

5. In figure the switch is closed at t = 0, with the capacitor of capacity $1\mu F$ and having initial charge of $20\mu C$ (the polarity shown). The left square loop has very large dimensions as compared to the distance between loops and a resistance of 10Ω is connected in series with capacitor as shown. A wire of length 8 mm and resistance per unit length of $0.5\Omega/mm$ is bent in the form of square loop 'A' and placed at a distance x = 1 m from left loop.



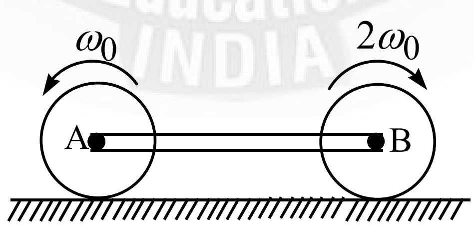
A) The mutual inductance of the system will be $8 \times 10^{-13} H$

B) The induced current in the loop A will be clockwise

C) The induced current in the loop A will be anticlockwise

D) The induced current in the loop A at time $t = 2RC\ell n2$ will be $10^{-8} A$

6. Two identical disc having radius R are rotating in opposite sense are joined with light rod having frictionless groove A and B and placed on rough horizontal surface as shown in figure. Initially translation velocity of each disc is zero. Then



A) Final velocity of B is $\dfrac{R\omega_0}{6}$

B) Rod's length will increase initially due to stress

C) Finally rod has tensile stress

D) Mechanical energy will decrease

SECTION-2(Maximum Marks: 12)

Paragraph with Numerical

- This section contains THREE (03) question stems.
- There are TWO (02) questions corresponding to each question stem.
- The answer to each question is a NUMERICAL VALUE.
- For each question, enter the correct numerical value corresponding to the answer in the designated place using the mouse and the on-screen virtual numeric keypad.
- If the numerical value has more than two decimal places, truncate/round-off the value to TWO decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +2 If ONLY the correct numerical value is entered at the designated place;
Zero Marks: 0 In all other cases.

Question Stem for Question Nos. 7 and 8

Question Stem

Experiments are going on to produce large scale energy by using Deuterium $\left({}_1H^2\right)$ and Tritium $\left({}_1H^3\right)$. They are using laser beams to compress the material. Fuel pellets are made by mixing equal number of atoms of ${}_1H^3$ and ${}_1H^2$. Such fuel pellets are spheres of radius $r = 200\mu m$ and its density is $200\ kg/m^3$. When laser beams hit the pellet, it gets compressed and its density is increased by 1000 times. Assume that only 10% of the atoms take part in fusion reaction

$${}_1H^2 + {}_1H^3 \rightarrow {}_2He^4 + {}_0n^1 + Q$$

mass of ${}_1H^2 = 2.0141\ u$

mass of ${}_1H^3 = 3.0160\ u$

mass of ${}_2H^4 = 4.0026\ u$

mass of ${}_0n^1 = 1.0087\ u$

(take, $1\ amu = 931\ MeV/c^2$)

7. The Q-value for above reaction in Mev is ________

8. The energy produced in laser action of a pellet is 10^x J. the value of x is _______

Question Stem for Question Nos. 9 and 10

Question Stem

Two artificial satellites A and B of masses $m_A = 300$ kg and $m_B = 3000$ kg are orbiting in circular orbit around the earth as shown in the figure. Earth is at rest. The initial orbit radius of A and B is r = 9000 km. Satellite B is ahead of A by a time ΔT. A rocket is fired from satellite 'A' in such a way that its speed decreases instantly by 0.6% as a result it moves in elliptical orbit as shown in figure-2. The new path traced by A is in a way such that it coincides with B at the initial position of A in one rotation as shown in figure-3. (Assuming Newton's law of gravitation and Kepler's law is valid for satellite A and B)

(π=3.14, G=6.67×10^{-11} S.I unit, mass of earth = Me = 6×10^{24} kg and radius of earth = R_e = 6400 km)

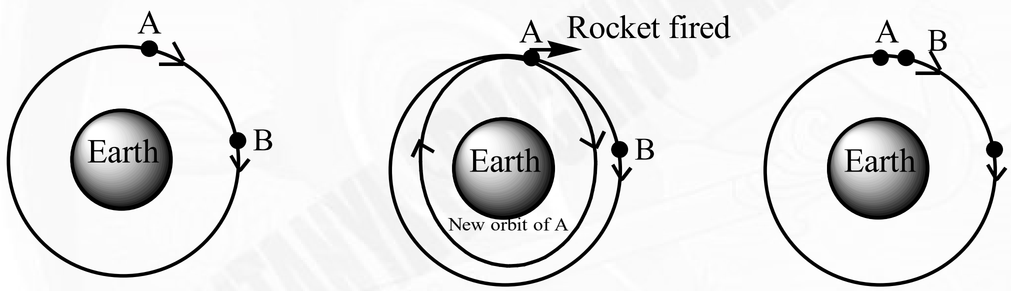


Figure-1 Figure-2 Figure-3

9. The new semi major axis of satellite A in km is ________

10. The value ΔT in seconds is ________

Question Stem for Question Nos. 11 and 12

Question Stem

Incident wave $y = A\sin\left(ax + bt + \frac{\pi}{2}\right)$ is reflected by an obstacle at x = 0 which reduces intensity of reflected wave by 36%. Due to superposition, the resulting wave consists of a standing wave and a traveling wave formed which is given by

$y = -1.6A\sin ax.\sin bt + cA\cos(bt + ax)$

Where A, a, b, c are positive constants.

In this resultant wave maximum speed of particle is V.

11. Ratio of amplitude of reflected wave to incident wave is________

12. Value of c is ________

SECTION-3(Maximum Marks: 12)

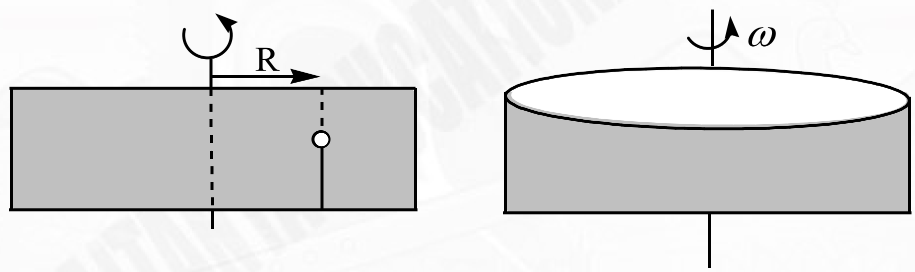
Paragraph with Single Answer Type

- This section contains TWO (02) paragraphs. Based on each paragraph, there are TWO (02) questions.
- Each question has FOUR options (A), (B), (C) and (D). ONLY ONE of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +3 If ONLY the correct option is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: −1 In all other cases.

Paragraph-I

A closed cylindrical container with a vertical axis is completely filled with liquid (ρ_ω). A plastic bead of density ρ_{body} and volume V is placed at distance R from the axis and is anchored to the bottom of the container by a thin thread of length ℓ. If as a result of the containers revolutions, the beads sink by h and final distance of bead from axis is r, (In the final state the total content of the container rotates at the same angular speed) then answer the following questions



13. What is force exerted by liquid on the bead?

A) $\rho_\omega V g$ **B)** $(\rho_\omega - \rho_{body})V g$

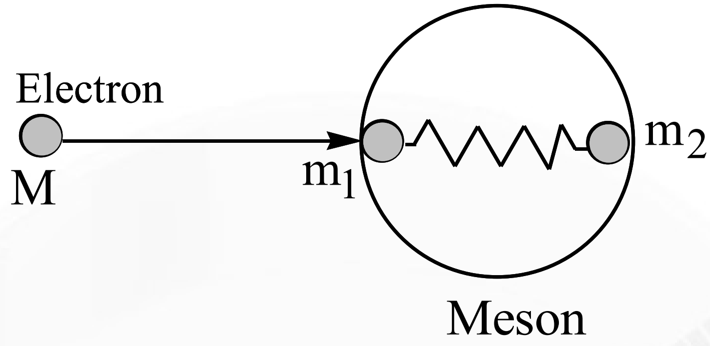
C) $(\rho_\omega - \rho_{body})V\sqrt{g^2+(\omega^2 r)^2}$ **D)** $\rho_\omega V\sqrt{g^2+(\omega^2 r)^2}$

14. The value of ω is :

A) $\omega=\sqrt{\dfrac{(R+r)g}{(\ell+h)r}}$ **B)** $\omega=\sqrt{\dfrac{(R-r)g}{(\ell-h)r}}$ **C)** $\omega=\sqrt{\dfrac{(R+r)g}{(\ell-h)r}}$ **D)** $\omega=\sqrt{\dfrac{(R-r)g}{(\ell+h)r}}$

Paragraph-II

Meson is composed of two quarks and the interaction between the quarks is complicated. Research of meson can be done by studying the inelastic collisions between the meson and high energy electrons. As the collision is quite complicated, scientists invented a simplified model called "parton model" to grasp the main content during collision. In the model, the electron first collides with part of meson (e.g. one of the quarks) elastically. Then the energy and momentum are transferred to the other quark and thus the whole meson during subsequent interaction. This simplified model is described by the following:



An electron of mass M and energy E collides with a quark of mass m_1 in a meson. The other quark in the meson has mass m_2. The quark are connected by a massless spring of natural length L which is at equilibrium before collision. All movements are on a straight line and neglect the effect of relativity. Find, after collision,

15. The energy gain of the quark m_1 is

A) $\dfrac{4Mm_1}{(M+m_1)^2}E$ B) $\dfrac{2Mm_1}{(M+m_1)^2}E$ C) $\dfrac{Mm_1}{(M+m_1)^2}E$ D) $\dfrac{3Mm_1}{2(M+m_1)^2}E$

16. The minimum kinetic energy of the meson as a whole system is

A) $\dfrac{4Mm_1^2}{(M+m_1)^2(m_1+m_2)}E$ B) $\dfrac{4Mm_1^2}{(M+m_2)^2(m_1+m_2)}E$

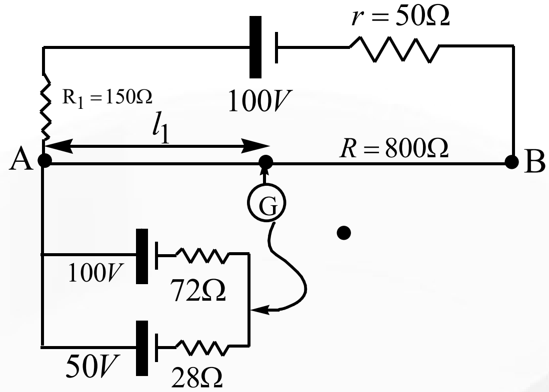
C) $\dfrac{2Mm_1^2}{(M+m_2)^2(m_1+m_2)}E$ D) $\dfrac{2Mm_1^2}{3(M+m_2)^2(m_1+m_2)}E$

SECTION-4(Maximum Marks: 12)
Non-Negative Integer Answer Type

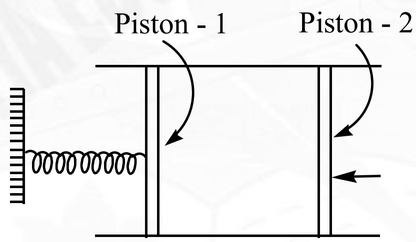
- This section contains THREE (03) questions.
- The answer to each question is a NON-NEGATIVE INTEGER.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4 If ONLY the correct integer is entered;
Zero Marks: 0 In all other cases.

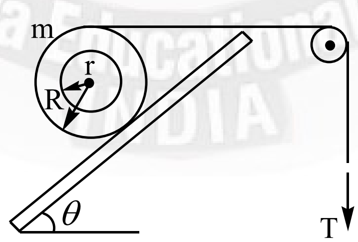
17. In the circuit shown length of AB is 100 cm. Then find the value of ℓ_1 (in cm). Where ℓ_1 corresponds to null point. Resistance of wire AB is 800Ω



18. A long container has air enclosed inside at room temperature and atmospheric pressure (10^5pa). It has a volume of 20,000 cc. The area of cross section is $100\ \text{cm}^2$ and force constant of spring is $k_{spring} = 1000\text{N/m}$. We push the right piston isothermally and slowly till it reaches the original position of the left piston which is movable. What is the final length of air column in m. Assume that spring is initially relaxed.



19. A spool, initially at rest, is kept on a frictionless incline making an angle $\theta = 37^\circ$ with the horizontal. The mass of the spool is 2kg and it is pulled by a string as shown with a force of T = 10N. The string connecting the spool and the pulley is initially horizontal. Find the initial acceleration of the spool on the incline. Express your answer in m/s^2.



CHEMISTRY

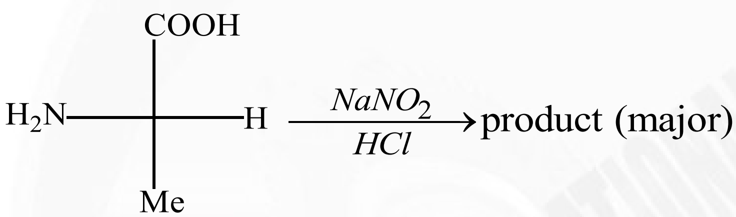
Max. Marks: 60

SECTION-1(Maximum Marks: 24)

One or More Type

- This section contains SIX (06) questions.
- Each question has FOUR options (A), (B), (C) and (D). ONE OR MORE THAN ONE of these four option(s)
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks: +4 If only (all) the correct option(s) is(are) chosen;
  Partial Marks: +3 If all the four options are correct but ONLY three options are chosen;
  Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
  Partial Marks: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
  Zero Marks: 0 If unanswered;
  Negative Marks: −2 In all other cases.

20.

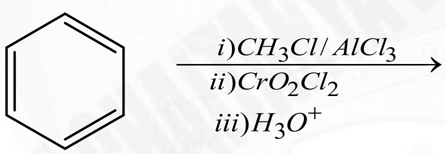


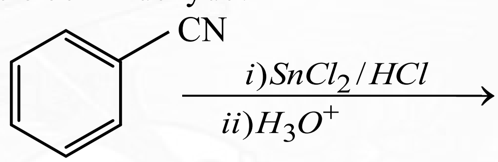
Select correct statement(s).

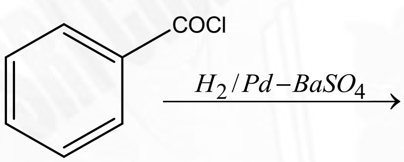
A) Major product is L-lactic acid. B) Reactant is called as D-alanine.

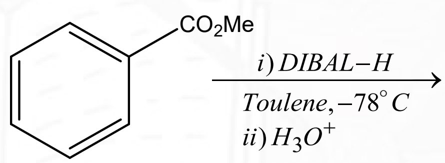
C) Reaction involves diazotization. D) Product on heating gives lactide.

21. Which of the following reaction(s) products benzaldehyde?

A) 

B) 

C) 

D) 

22. Which of the following statements is/are CORRECT?

A) 1 mole Benzene $\left(P^{\circ}_{Benzene} = 42mm\right)$ and 2 mole Toluene $\left(P^{\circ}_{Toluene} = 36mm\right)$ liquid mixture will have mole fraction of benzene in vapour phase equal to 7/19.

B) If two liquids that form an ideal solution are mixed, the change in entropy is positive.

C) Higher the value of K_H (Henry's law constant) at a given pressure and temperature , lower is the solubility of the gas in the liquid.

D) Solubility of a gas in liquid increase with increase in temperature at a given pressure

23. Which of the following is/are correct for the vander waal's constant a and b

A) a is the measure of force of attraction in between the particles.

B) b is the excluded volume for 1-mole gas.

C) Higher is the value of a easier is the liquification of the gas.

D) Lower value of b indicates that the gas is more compressible.

24. Which of the following statement(s) is/are correct?

A) Pyrophosphoric acid has only two $P-OH$, two $P=O$ and one $P-O-P$ bonds

B) First ionization energy of $O^{2-}(g)$ is exothermic

C) Both Hg^{2+} and Pb^{2+} gives red and yellow precipitates respectively by adding excess of KI solution

D) Among $\left[Fe(H_2O)_6\right]^{3+}$, $\left[Fe(NH_3)_6\right]^{3+}$, $\left[Fe(C_2O_4)_3\right]^{3-}$ and $[FeCl_6]^{3-}$; $\left[Fe(C_2O_4)_3\right]^{3-}$ is the most stable complex

25. Which of the following statement(s) is/are incorrect?

A) O^{2-}, F^-, Na^+ and Mg^{2+} are all isoelectronic and have the same nuclear charge

B) In group 13, the stability of +1 oxidation state increases down the group

C) The atomic size of gallium is greater than that of aluminium

D) Aluminium dissolves in dilute HCl and liberates H_2 but conc HNO_3 renders aluminium passive by forming a protective oxide layer on the surface

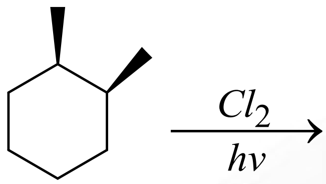
SECTION-2(Maximum Marks: 12)

Paragraph with Numerical

- This section contains THREE (03) question stems.
- There are TWO (02) questions corresponding to each question stem.
- The answer to each question is a NUMERICAL VALUE.
- For each question, enter the correct numerical value corresponding to the answer in the designated place using the mouse and the on-screen virtual numeric keypad.
- If the numerical value has more than two decimal places, truncate/round-off the value to TWO decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks: +2 If ONLY the correct numerical value is entered at the designated place;
  Zero Marks: 0 In all other cases.

Question Stem for Question Nos. 26 and 27

Question Stem



(Consider only mono chloro products)

If X is the number of 2^0 mono chloro products and Y is the number of fractions obtained upon fractional distillation of the product mixture

26. Find the value of X (including stereo)?

27. Find the value of Y ?

Question Stem for Question Nos. 28 and 29

Question Stem

The molar conductivity of NaCl varies with the concentration as shown in the following table and all values follows the equation

$\lambda_m^C = \lambda_m^\infty - b\sqrt{C}$ Where λ_m^C = molar conductivity at given concentration

λ_m^∞ =Molar conductivity at infinite dilution

C =Molar concentration

Molar Concentration of NaCl	Molar conductivity $\left[\lambda_m^c\right]$ $ohm^{-1}cm^2mole^{-1}$
4×10^{-4}	107
9×10^{-4}	97
16×10^{-4}	87

When a certain conductivity cell was filled with $25\times10^{-4}\,M\ NaCl$ solution. The resistance of the cell was found to be 1000 ohm.

28. What is the molar conductivity of NaCl at infinite dilution in $ohm^{-1}cm^2mol^{-1}$ $u$int ?

29. What is the cell constant of the given conductivity cell in cm^{-1} $unit$

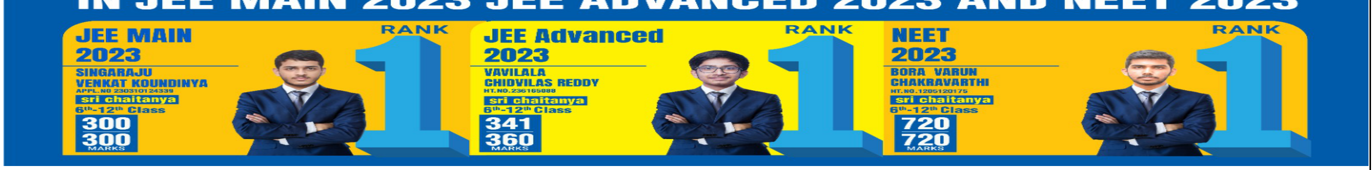
Question Stem for Question Nos. 30 and 31

Question Stem

Aqueous cupric sulphate solution(blue in colour) gives:

(i) a green precipitate (X) with aqueous KF and

(ii) a bright green solution (Y) with aqueous KCl



30. Co-ordination number of central metal ion in the complex ion of X is___

31. Co-ordination number of central metal ion in complex ion of Y is_____

SECTION-3(Maximum Marks: 12)

Paragraph with Single Answer Type

- This section contains TWO (02) paragraphs. Based on each paragraph, there are TWO (02) questions.
- Each question has FOUR options (A), (B), (C) and (D). ONLY ONE of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +3 If ONLY the correct option is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: −1 In all other cases.

Paragraph-I:

An optically active compound (A) with molecular formula $C_5H_{13}N$ was reacted with excess of methyl iodide to give a compound (B) which on heating with $AgOH$ gave a compound (C), compound (C) on heating with alkaline $KMnO_4$ gave isobutyric acid and CO_2. The compound (A) on reaction with HNO_2 gave an alcohol (D).

32. A compound (A) is

A) $CH_3-CH_2-CH_2-\underset{\large|\atop NH_2}{CH}-CH_3$

B) $(CH_3)_2CH-\underset{\large|\atop NH_2}{CH}-CH_3$

C) $CH_3-CH_2-\underset{\large|\atop NH-CH_3}{CH}-CH_3$

D) $(CH_3)_2CH-CH_2-NH-CH_3$

33. The number of possible tertiary amines of the given compound (A) are

A) 4 B) 3 C) 6 D) 5

Paragraph-II:

An ideal solution is prepared at 25^0C by dissolving 0.4kg of solid AB (having rock-salt structure, B^- ions at corner) and 5 moles liquid 'P' in 20 moles liquid 'Q' .The solid is non-volatile and nonelectrolyte in the solution .At 25^0C, The vapor pressure of pure liquids 'P' and 'Q' are 100 torr and 200 torr respectively and the vapor pressure of solution is 150 torr. The shortest distance between two A^+ ions in the solid is $200\sqrt{2}\ pm\left(N_A = 6\times10^{23}\right)$

34. The molar mass of solid AB is:

A)100 gm/mol **B)** 80 gm/mol **C)** 40 gm/mol **D)** 160 gm/mol

35. The density of solid AB is:

A) $\frac{25}{3}gm/cm^3$ B) $\frac{50}{3}gm/cm^3$ C) $\frac{25}{6}gm/cm^3$ D) $\frac{25}{8}gm/cm^3$

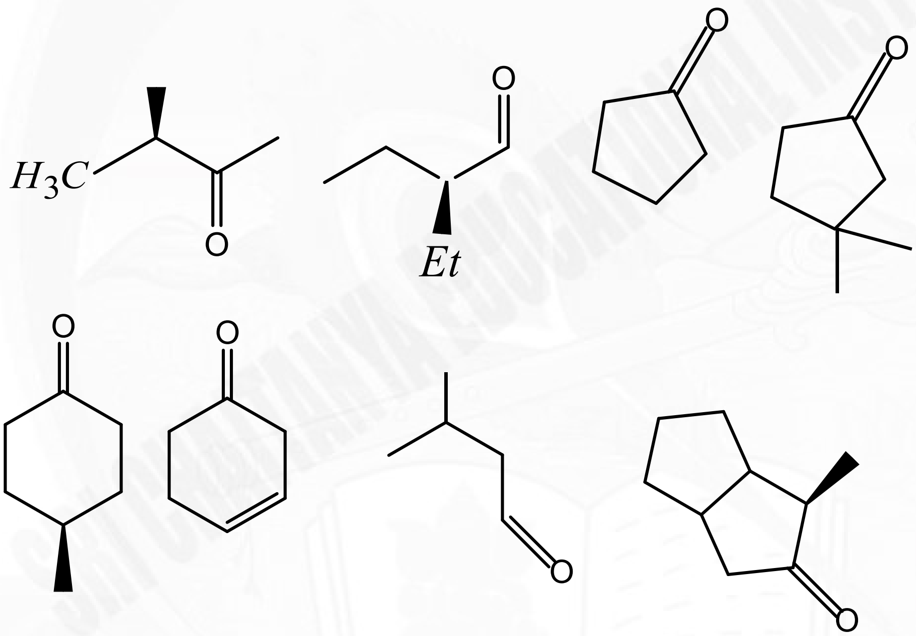
SECTION-4(Maximum Marks: 12)

Non-Negative Integer Answer Type

- This section contains THREE (03) questions.
- The answer to each question is a NON-NEGATIVE INTEGER.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4 If ONLY the correct integer is entered;
Zero Marks: 0 In all other cases.

36. How many of the following compounds give racemic mixture with CH_3MgCl.



37. Ratio of 'σ' bonds to 'π' bonds in $P_4O_{12}^{4-}$ is:

38. Fusion of chromite ore with sodium carbonate in free excess of air gives two compounds X and Y(contains transition metals) along with CO_2.
Sum of the spin – only magnetic moments (nearest integer)of the transition metal ions in X and Y is____
(Given that: Among X and Y , if any compound contains two or more than two transition metals, then consider average oxidation state for calculating spin – only magnetic moments)
(Given that: $\sqrt{2}=1.4$, $\sqrt{3}=1.7$, $\sqrt{5}=2.2$, $\sqrt{7}=2.6$, $\sqrt{8}=2.8$, $\sqrt{35}=5.9$)

MATHEMATICS

Max. Marks: 60

SECTION-1(Maximum Marks: 24)

One or More Type

- This section contains SIX (06) questions.
- Each question has FOUR options (A), (B), (C) and (D). ONE OR MORE THAN ONE of these four option(s)
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks: +4 If only (all) the correct option(s) is(are) chosen;
  Partial Marks: +3 If all the four options are correct but ONLY three options are chosen;
  Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
  Partial Marks: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
  Zero Marks: 0 If unanswered;
  Negative Marks: −2 In all other cases.

39. Let [.] denotes the greatest integer function, $f:(0,1)-\left\{\frac{1}{3}\right\}\to R$ be the function defined as $f(x)=[4x]\left(x-\frac{1-[x]}{3}\right)^{[2x]}\left(x-\frac{[2x]+1}{4}\right)^{3+[x]}$. Then which of the following options is/are correct?

A) The function f is discontinuous at exactly two points in its domain

B) There exists exactly two points in $(0,1)$ at which the function f is continuous but NOT differentiable

C) The function f is NOT differentiable at more than three points in $(0,1)$

D) The minimum value of f is 0

40. If $a,b,c\in R^{+}$ and $a^2bc+4abc+4bc=81$ such that $(2a+b+c+4)$ assumes its least value, then select the correct alternative/s

A) The value of $(a+b+c)$ is a prime number

B) The value of abc is a prime number

C) $b=3a$

D) $b\le c$

41. Given points A,B,C with their position vectors as below,

$\vec{a}=-2024\vec{i}+2025\vec{j}+2022\vec{k}$,

$\vec{b}=2^{2022}3^{2023}\vec{i}+3^{2021}5^{2025}\vec{j}-\vec{k}$,

then which of the following options is/are correct?(where O is origin)

A) The point $C(\vec{c})=\frac{2023}{2024}\vec{a}+\frac{1}{4048}\vec{b}$ lies inside of triangle formed by O, A, B as vertices.

B) The point $C(\vec{c})=\frac{2024}{2025}\vec{a}+\frac{3}{2025}\vec{b}$ lies outside of triangle formed by O, A, B as vertices, but lies inside of $\angle AOB$

C) The point $C(\vec{c})=\frac{-2}{x^2+1}\vec{a}+\frac{x^2-1}{x^2+1}\vec{b}$ lies outside of triangle formed by O, A, B as vertices, but lies inside of $\angle OAB\ \forall x\in R$

D) The point $C(\vec{c})=\frac{2}{x^2+1}\vec{a}-\frac{x^2}{x^2+1}\vec{b}$ lies outside of triangle formed by O, A, B as vertices, but lies inside of $\angle OBA\ \forall x\in R$

42. Let $f:\mathbb{R}\to\mathbb{R}$ be a continuous function such that $f(x+1)=\frac{1}{2}f(x)$ for all $x\in\mathbb{R}$, and let $a_n=\int_0^n f(x)dx$ for all integers $n\geq 1$. Then, which of the following is/are CORRECT?

A) $\lim_{n\to\infty} a_n$ exists and equals $\int_0^1 f(x)dx$.

B) $\lim_{n\to\infty} a_n$ does not exist.

C) $\lim_{n\to\infty} a_n$ exists if and only if $\left|\int_0^1 f(x)dx\right|<1$.

D) $\lim_{n\to\infty} a_n$ exists and equals $2\int_0^1 f(x)dx$.

43. Let

$$S_k=3^k\ {}^{100}C_0\ {}^{100}C_k-3^{k-1}\ {}^{100}C_1\ {}^{99}C_{k-1}+3^{k-2}\ {}^{100}C_2\ {}^{98}C_{k-2}+...+(-1)^k\ {}^{100}C_k\ {}^{100-k}C_0,$$

then select correct alternative(s) $(k=0,1,2,.....,100)$

A) S_{67} is the greatest amongst these 101 numbers

B) $S_1+S_3+S_5+....+S_{99}\leq S_2+S_4+....+S_{100}$

C) S_{66} is the greatest amongst these 101 numbers

D) $S_1+S_3+S_5+....+S_{99}>S_2+S_4+....+S_{100}$

44. $\prod_{i=1}^{n} P_i^{a_i \cos^2 a_i} = 2023$, where P_i represents the i^{th} prime number (i.e., for example $P_1 = 2, P_2 = 3, P_3 = 5.......$) such that $P_n < 2023$ and $a_i \in R$. If elements of set A is the solution of the given equation represented by ordered n-tuples $(a_1, a_2, a_3......a_n)$, then which of the following is/are INCORRECT?

A) Number of elements in set A is finite

B) If $a_i \le 9 \ \forall \ i = 1,2,....,$ n, then the number of element in set A is 25

C) If $a_i \le 0 \ \forall \ i = 1,2,....,$ n, then A is null set

D) If $a_i \le 9 \ \forall \ i = 1,2,....,$ n, then A in singleton set

SECTION-2(Maximum Marks: 12)

Paragraph with Numerical

- This section contains THREE (03) question stems.
- There are TWO (02) questions corresponding to each question stem.
- The answer to each question is a NUMERICAL VALUE.
- For each question, enter the correct numerical value corresponding to the answer in the designated place using the mouse and the on-screen virtual numeric keypad.
- If the numerical value has more than two decimal places, truncate/round-off the value to TWO decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +2 If ONLY the correct numerical value is entered at the designated place;
Zero Marks: 0 In all other cases.

Question Stem for Question Nos. 45 and 46

Question Stem

Consider set $S = \left\{(a,b) : (a+3)t^2 = 3 - a \text{ and } bt^2 - 4t + b = 0\right\}$, where t is are real parameter. Let C is curve which is formed by all elements of set S where (a,b) is a point in R^2. Tangents are drawn from the point $P(3,4)$ to the curve C touching the curve C at point Q and R.

45. The circumcentre of triangle PQR is (α, β), then the value of $\alpha + 3\beta$ is ___

46. Curve E is equation of ellipse whose foci are Q,R and curve E is passing through point P, then eccentricity of ellipse E is e, Then the value of $\left(\frac{e}{3-\sqrt{5}}\right)^2$ is ___

Question Stem for Question Nos. 47 and 48

Question Stem

Let $L_1 : 4x - 3y + 13 = 0, L_2 : 4x - 3y = 37, L_3 : 3x + 4y = 34$ are three lines in xy plane and $L_4 : (1+\lambda)x + (1-\lambda)y = 24$ is a variable line. $P(a,b)$ is centre of circle which touches lines L_1, L_2 and L_3.

On the basis of above information, answer the following questions:

47. Maximum value of $a+b$ is ___

48. If L_1, L_2, L_3 and L_4 form a quadrilateral, then the value of λ for which slope of line L_4 takes least positive integral value is ___

Question Stem for Question Nos. 49 and 50

Question Stem

Given set of vectors:

$$\vec{a} = \frac{-2}{3}t^3\vec{i} - \frac{3}{2}t^2\vec{j} + \left(\frac{\lambda}{3} - 10\right)\vec{k},$$

$$\vec{b} = \vec{i} + 2\vec{j} + 3\vec{k},$$

$$\vec{c} = 4\vec{i} + 5\vec{j} + 6\vec{k},$$

t is a real number which satisfies such that given three vectors are linearly dependent vectors, and $t = x + \dfrac{1}{x}$ and $\lambda \in R$ then

49. The value of λ, for which the given equation will have three real and distinct solutions of x is ___

50. If the given equation has exactly two real and distinct solution(s) of x, then the number of integral values of λ in $[-25, 15]$ is ___

SECTION-3(Maximum Marks: 12)

Paragraph with Single Answer Type

- This section contains TWO (02) paragraphs. Based on each paragraph, there are TWO (02) questions.
- Each question has FOUR options (A), (B), (C) and (D). ONLY ONE of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +3 If ONLY the correct option is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: −1 In all other cases.

Paragraph-I:

Let $\displaystyle\int \frac{6x^3 + 2x^2 + 8x + 1}{\sqrt{1+x^2}}dx = \left(ax^2 + bx + c\right)\sqrt{1+x^2} + p\ln\left|x + \sqrt{1+x^2}\right| + k,$

where $a, b, c, p \in R$ and k is constant of integration. Column I contains the variables a, b, c, p, Column II contains their values and Column III contains a combination of the variables a, b, c, p

Column I		Column II		Column III	
A)	a	P)	0	I)	$c+p$
B)	b	Q)	4	II)	$p+2b$
C)	c	R)	2	III)	$c-2a$
D)	p	S)	1	IV)	$2a$

51. The function $f:\{A,B,C,D\}\rightarrow\{P,Q,R,S\}$ relating the values of the variables in column I to that in column II (For e.g. suppose the values of a, b are 2 and 1 respectively, then f(A)=R, f(B)= S etc) is

A) Many-one Into

B) One-one Onto

C) Not a function as some element/s of Column I have multiple values in Column II

D) Not a function as some element/s of Column I do not have a value in Column II

52. The function $g:\{I, II, III, IV\}\rightarrow\{P, Q, R, S\}$ relating the values of the variables in Column III to that in Column II (For e.g. suppose the values of c-2a, 2a are 0 and 0 respectively, then f (III) = P, f (IV) = P etc) is

A) Many-one Into

B) One-one Onto

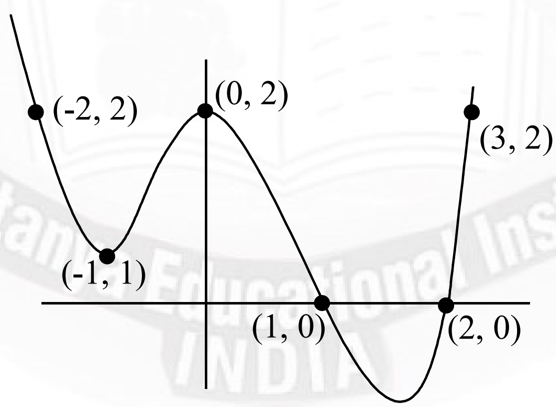
C) Not a function as some element/s of Column 3 have multiple values in Column II

D) Not a function as some element/s of Column 3 do not have a value in Column II

Paragraph-II:

The graph of the polynomial

$y=p(x)=x^n+a_1x^{n-1}+a_2x^{n-2}+\ldots+a_n, a_i\in R$ is given. It has local extremas at x = -1 and 0 and no other critical point other than visible in the graph.



53. Let $S=\{4,5,7,10\}$,then how many value(s) of S ,which are not possible for n?

A) 1 **B)** 2 **C)** 3 **D)** 4

54. Given four statements:

Statement-I: $\left.\dfrac{d^3y}{dx^3}\right|_{x=0} = 0$

Statement-II: P(x) = 0 has at least 4 non real roots

Statement-III: $\left.\dfrac{d^3y}{dx^3}\right|_{x=0} \neq 0$

Statement-IV: P(x) = 0 has at least 6 non real roots

If $a_{n-2} = 0$, then the number of statement(s) from above must be true?

A) 1 **B)** 2 **C)** 3 **D)** 4

SECTION-4(Maximum Marks: 12)

Non-Negative Integer Answer Type

- This section contains THREE (03) questions.
- The answer to each question is a NON-NEGATIVE INTEGER.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4 If ONLY the correct integer is entered;

Zero Marks: 0 In all other cases.

55. The units digit in the integral part of the expansion of $\left(15+\sqrt{220}\right)^{19}\left(1+\left(15+\sqrt{220}\right)^{63}\right)$ is ___

(Here integral part means the greatest integer less than or equal to that number)

56. Let α be a 6^{th} root of unity, and define $S = \left\{\alpha^n + \alpha^{-n} : n \in I\right\}$, then the sum of all distinct possible values of n(S) is ___(Here n(S) denotes the number of distinct elements of the set S)

57. If $A_i\left(i = 1, 2, ..., n\right)$ represents all possible matrices satisfying $A_i^2 = \begin{bmatrix} 1 & 2 \\ 0 & 4 \end{bmatrix}$, and let x_i denote the absolute value of determinant of A_i then the value of $\sum_{i=1}^{n}\left(x_i - 1\right)$ is equal to

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